

Nikhilesh Vijay Chirutala

Software Engineer | Python Backend & API Engineer

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SUMMARY

Python Backend and API Engineer with 3 years of experience developing and maintaining production applications, backend services, REST APIs, and data-driven workflows across cloud, CRM, AI, and contact-center platforms. Experienced in Python, FastAPI, REST API development, relational databases, MongoDB, SQL, AWS, Selenium-Python testing, and backend automation. Strong experience working with production data workflows, API integrations, validation, automated testing, troubleshooting, deployment activities, and Git-based development. Experienced in building reliable backend solutions across application, database, and cloud layers.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript (ES6+), TypeScript, Shell Scripting

Backend & API Development: Python, FastAPI, Flask, REST APIs, API Development, Backend Services, Backend Automation

Databases: MySQL, PostgreSQL, MongoDB, SQL, Relational Databases, Data Modeling, CRUD Operations

Testing & Automation: Selenium-Python, API Testing, Unit Testing, Jest, Postman, Automated Testing, Debugging, Test Automation

Cloud & Platforms: AWS, Amazon S3, AWS Lambda, Vercel, Linux, CI/CD

SCM & Development Tools: Git, GitHub, GitHub Actions, Postman, Code Reviews, SonarQube

Data & Integrations: Data Processing, Data Validation, Data Workflows, API Integration, WebSockets, Twilio, GitHub API

Security: Authentication, Authorization, RBAC, ACL, JWT

AI & Machine Learning: RAG, LLM Integration, Vector Search, Prompt Engineering

WORK EXPERIENCE

Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog, Grafana

- Develop and maintain production-grade Python backend services for a high-volume cloud contact center platform, supporting data processing, call-recording workflows, storage operations, and production use cases.
- Migrated a legacy Perl-based workflow to a long-running Python service deployed across 40 production clusters, supporting staged rollout, feature flags, validation, rollback, and production monitoring.
- Built Python service integrations with a Call Intelligence microservice, implementing APIs, eligibility checks, account and campaign controls, processing limits, retries, response validation, and result persistence.
- Developed validation and recovery workflows for production data, handling incomplete processing, failed operations, inconsistent storage states, and recovery scenarios across multiple environments.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage workflows.
- Supported production deployments across multiple clusters through configuration, staged validation, monitoring, troubleshooting, and rollback activities.
- Implemented production observability using Datadog, Grafana dashboards, and Sentry to monitor service health, diagnose failures, and support production incident resolution.
- Investigated and resolved production issues across Python services, APIs, databases, and cloud storage while improving reliability of recording-processing workflows.

Software Engineer • Mantra Technologies

Oct 2023 – Mar 2026 | Hyderabad, India

- Developed production backend services and REST APIs using Python and FastAPI, implementing business logic, validation, application workflows, and data-driven functionality.
- Worked with MongoDB and relational databases including PostgreSQL to implement data models, CRUD operations, relationships, validation, and database-backed application workflows.
- Integrated external services and APIs while implementing authentication, authorization, RBAC, ACL, and JWT-based access control across application workflows.
- Supported production releases through API testing, automated testing, debugging, root-cause analysis, deployment validation, code reviews, Git-based development, and defect resolution.

Key Projects:

CRM.io — AI-Driven CRM System

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM

- Developed backend modules and REST APIs using Python and FastAPI for Contacts, Tasks, Deals, Documents, communication, and account-management workflows.
- Implemented data-driven CRM workflows using MongoDB, including CRUD operations, global search, related records, validation, and account-level application functionality.

- Developed API integrations with Twilio and internal telephony services using WebSockets to support real-time CRM communication workflows.
- Implemented soft-delete and retention workflows using AWS Lambda and MongoDB, supporting automated record cleanup after the defined retention period.
- Built document-management workflows using Amazon S3, including browser-to-S3 uploads and storage of documents associated with CRM records.
- Built account-level reporting workflows and an MCP integration exposing CRM capabilities for AI-assisted retrieval and supported operations.

AI Bull — AI-Powered Trading & Investment Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed Python and FastAPI backend APIs supporting portfolio aggregation, investment tracking, screening, reporting, backtesting, and AI-powered workflows.
- Designed data-driven workflows for portfolio positions, stocks, mutual funds, investment information, and aggregated account data using MongoDB.
- Implemented configurable stock and mutual fund screening workflows with filtering capabilities and detailed asset-level information.
- Designed and implemented a backtesting engine for evaluating trading strategies against historical market data and analyzing strategy performance.
- Implemented a Retrieval-Augmented Generation (RAG) chatbot combining portfolio and financial information to provide contextual responses to portfolio-related queries.

PERSONAL PROJECTS

Data Processing & API Automation — Python Data Processing and API Testing Project

Python, REST APIs, Selenium-Python, SQL, Shell Scripting

- Developed Python workflows for consuming REST APIs, validating responses, processing structured data, and storing application results in relational database tables.
- Implemented Selenium-Python automation workflows for validating web application functionality and executing repeatable regression test scenarios.
- Created Shell scripts to automate execution of Python processing and testing workflows in a Linux environment.
- Implemented data validation and error-handling workflows for identifying failed API responses, invalid records, and processing exceptions.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform supporting projects, tasks, and team workflows with persistent application data.
- Implemented authentication and user management with Clerk and database-backed application workflows using Neon DB.

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application using Next.js and Vercel.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur